

Homotopy Continuation in Economics: Finding Equilibria Under Competition

Kevin Phan*

May 3, 2026

Abstract

Computational algebraic geometry has several unique applications in a variety of fields. The field of economics is rife with nonlinear systems of equations that are done for optimization and equilibrium analysis. One clear application of algebraic geometry is game theory, and I will focus on the game-theoretic models between competing firms, which is within the field of industrial organization. With this presentation, I aim to introduce the audience to the basics of algebraic geometry, focusing on Homotopy Continuation methods. Then I will go over the common models in economics to show applications through the programming language Julia. I will walk through a simple example of the Bertrand Price Model, where two firms compete with varying strategies in order to maximize profit to motivate this field of research. In the end, I hope to show the basics of homotopy continuation with an endearing example that motivates future research.

*This was done as a seminar project (MATH 480) under Max Hill.

1 Stackelberg Duopoly Model

This is probably the *simplest* one that I'll do, just to introduce people to the subject.

Math isn't too crazy and you can actually just solve it by hand.

2 Introduction

2.1 What is a homotopy and algebraic geometry?

3 Bertrand Price Model

The bertrand price model is a game between two firms, where the objective is to maximize profit (or so i think).

Consider products x, y , with p_x, p_y being the price of the goods. We have customer segmentation, with three types (1, 2, and 3). Thus, we can estimate the demand curves of the three customer types according to each product. In total we should have six demand curves.

Let type 1 customers only desire good x , leaving y unwanted. Thus, $Dx1 = A - p_x, Dy1 = 0$.

For type 3 customers, let us assume the opposite and have them only prefer good y . Thus $Dx3 = 0, Dy3 = A - p_y$.

Type 2, is a mixture of both goods. We additionally assume that x, y are imperfect substitutes and have a constant elasticity of substitution and constant elasticity of demand for the bundle/composite good. Thus,

$$Dx2 = np_x^{-\sigma} (p_x^{1-\sigma} + p_y^{1-\sigma})^{\frac{\gamma-\sigma}{-1+\sigma}} \quad (1)$$

$$Dy2 = np_y^{-\sigma} (p_x^{1-\sigma} + p_y^{1-\sigma})^{\frac{\gamma-\sigma}{-1+\sigma}} \quad (2)$$

We also have the following variables

- σ , the elasticity of substitution between the goods
- γ , the elasticity of demand for the composite good/bundle $(q_1^{(\sigma-1)/\sigma} + q_2^{(\sigma-1)/\sigma})^{\sigma/(\sigma-1)}$.

The total demand of x and y are quite simple, it is simply the sum of all three demand curves for each good. Suppose m is the unit cost for the firm, where we can then calculate profit.

$$R_x = (p_x - m)Dx$$

$$R_y = (p_y - m)Dy$$

For the uninitiated, the variables Dx, Dy is the quantity demanded. In economics, the price is usually the input of the function.

We can let the parameters in the model be equal to something, for example: $\sigma = 3, \gamma = 2, n = 2700, m = 1, A = 50$. Plugging in these values, we can calculate the marginal profits of each good, and when $MR_x = MR_y$, we should have equilibrium prices be satisfied.

Blah blah blah, take the derivatives of each thing, let Z be some part of it so it's a simpler polynomial, use a solver.